Changelog: PerfAudit v1.9

Release Date: November 9, 2026 | Build Type: Production

[NEW] Game FPS Logging

- **ETW Frame Capture:** Added game FPS logging using Event Tracing for Windows (ETW) subscribed to DXGI Present start events (IDs 42 and 55). The tool does not inject into or hook the game process and only reads system ETW events.
- **Dual Target Dropdowns:** Added a second dropdown "Select Game (FPS)" with a "None" option. CPU sampling and game FPS logging can run independently or in parallel on a shared timebase.
- **Independent Logging:** If one monitored process exits during a run (e.g. the game closes), the other logger continues running until manually stopped.
- **Live FPS Display:** Added an "FPS: 0" status label next to the CPU label, updated once per second while recording.

[NEW] Excel (.xlsx) Report Export

- **Excel Workbook Output:** Replaced CSV export with a single Excel workbook (.xlsx) per run generated with EPPlus 4.5.3.3.
- **Sheet "CPU_Usage":** Preserves the 100ms sample log, millisecond timestamps (' HH:mm:ss.fff), burst threshold indicators, and summary rows from the previous format.
- **Sheet "CPU_Graph":** Summary block (Average CPU %, Peak CPU %, Threshold used %, Total burst samples, Seconds containing a burst, Duration [mm]:ss), per-second peak CPU table, and line chart with peak CPU % per second, threshold line, and burst markers (empty seconds plotted as gaps).
- **Sheet "FPS_Log":** Per-frame log with timestamp, game process name, elapsed seconds, frametime (ms), and rolling 1-second FPS.
- **Sheet "FPS_Graph":** Summary stats (Average FPS, 1% low FPS (percentile), 0.1% low FPS (percentile), Lowest 1-second FPS, Worst frame (ms), Frames over 50 ms, Frames over 100 ms, Duration [mm]:ss), worst 10 frames table, per-second table, and charts for FPS and worst frametime per second.

[NEW] Type-Ahead Search in Dropdowns

- **Shared Prefix Search:** Added type-ahead search to both "Select Process (CPU)" and "Select Game (FPS)" dropdowns using a shared helper.
- **Incremental Typing:** Typing characters jumps to and selects the first matching item in list order by displayed text without resetting between keystrokes.
- **Keyboard Controls:** Backspace removes the last character and re-searches; Escape clears the search buffer. An automatic 1.5-second inactivity timeout resets the buffer, and the buffer clears on list refresh or focus loss while leaving arrow keys and mouse selection unchanged.

[REQUIREMENTS & DEPENDENCIES]

- **Administrator Rights:** The user must launch PerfAudit as administrator for FPS logging (ETW session creation requires administrative privileges; app.manifest is unchanged).
- **Dependencies:** Added EPPlus 4.5.3.3 (LGPL 2.1) and Microsoft.Diagnostics.Tracing.TraceEvent 3.1.8.